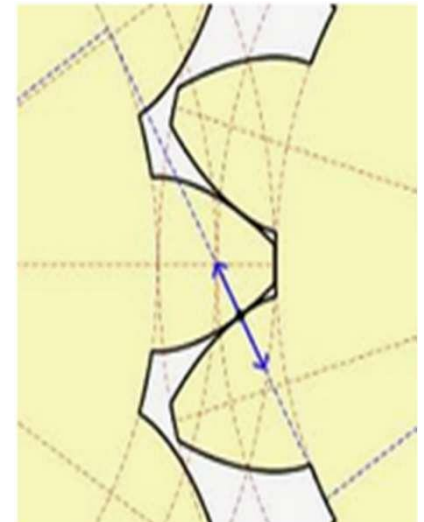


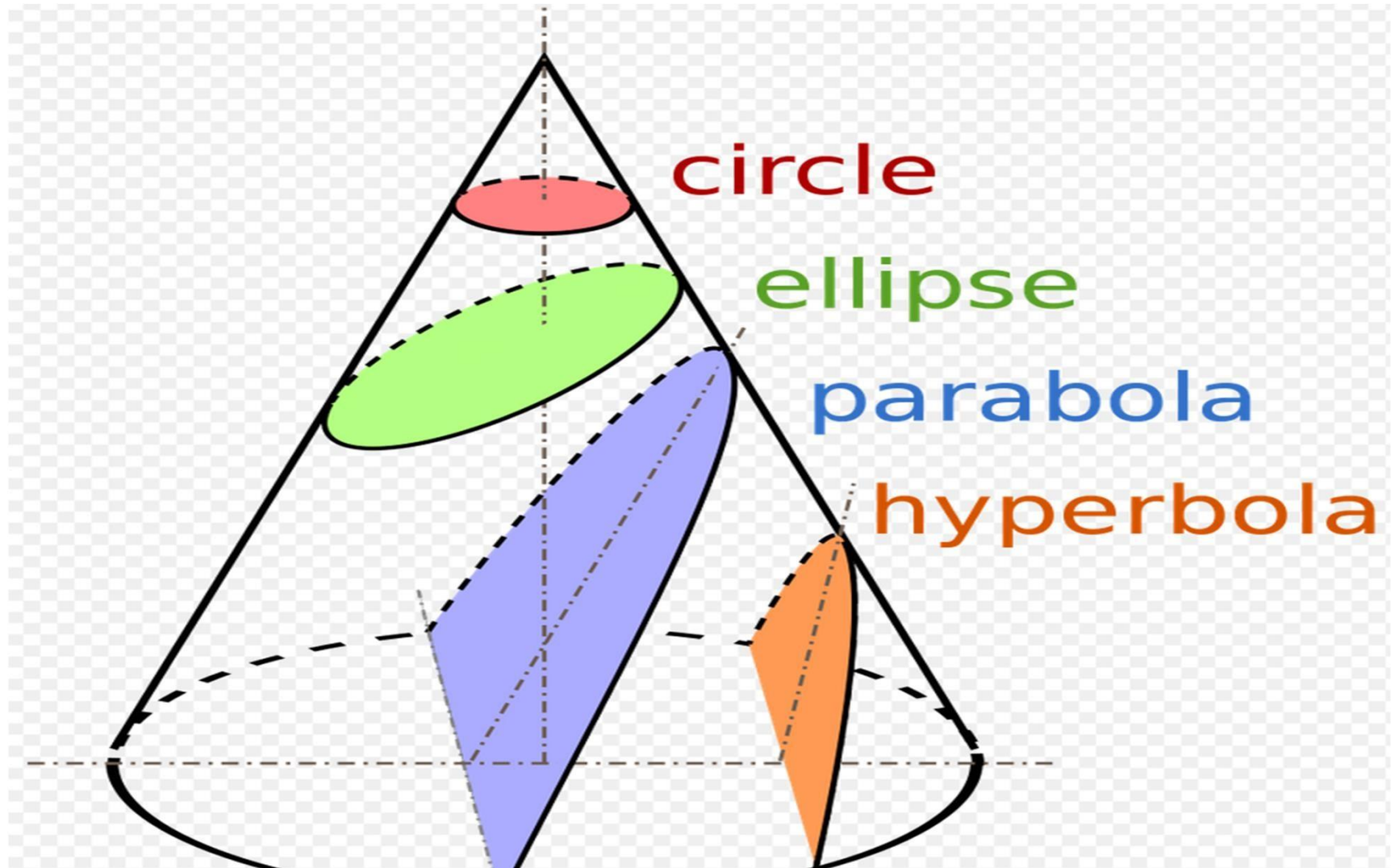
CURVES USED IN ENGINEERING GRAPHICS

Introduction

- Where curves are

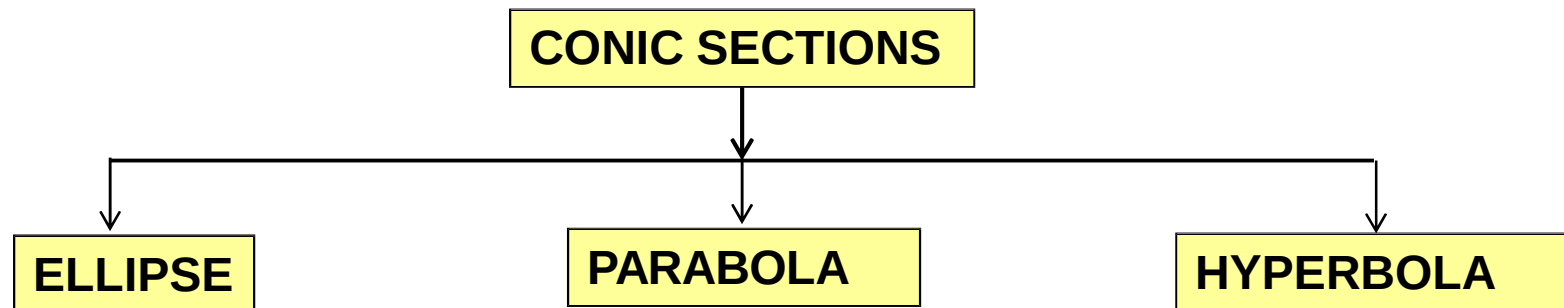


Conic

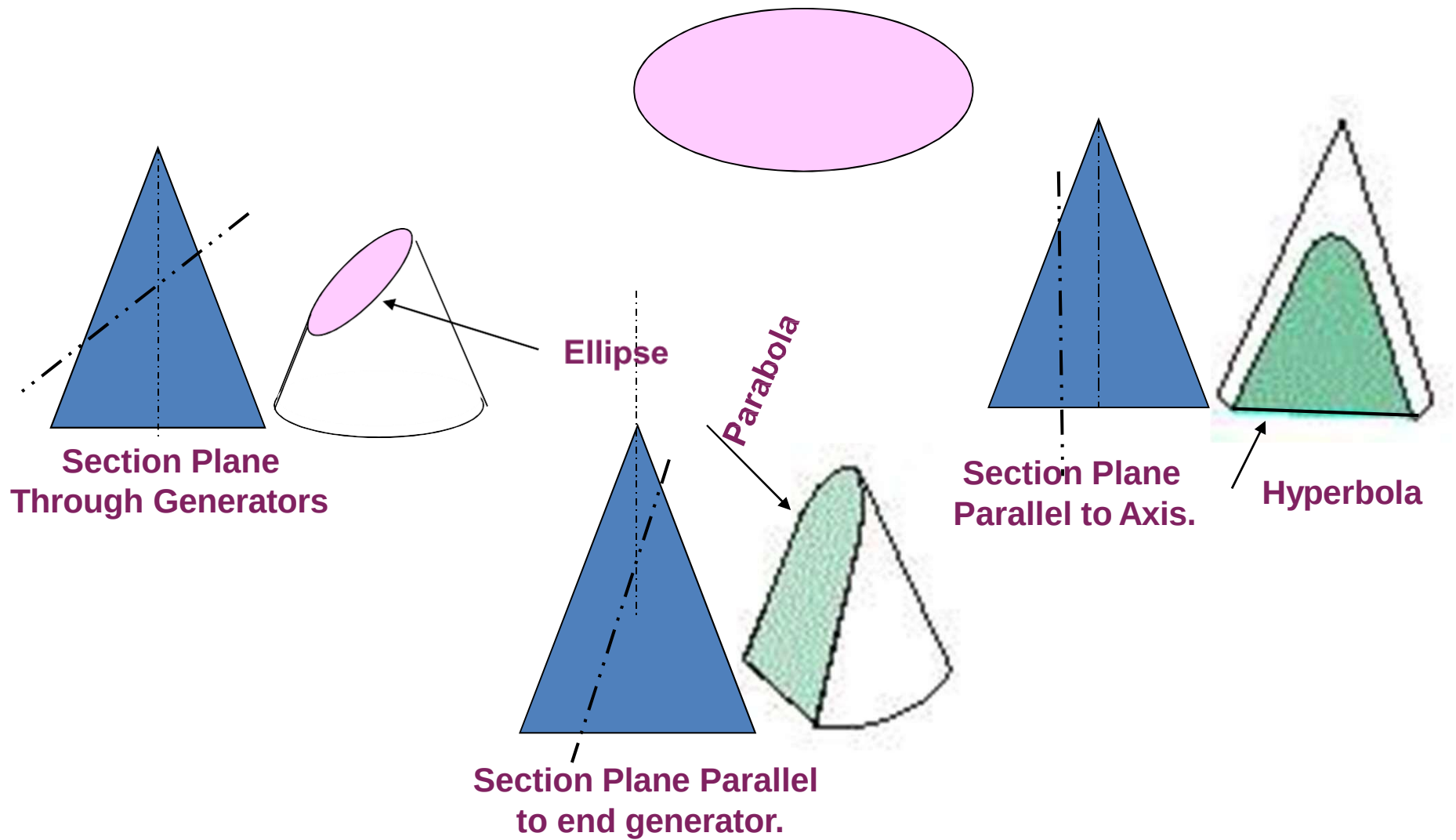


Conic sections

- **Conic section** (or simply **conic**) is a curve obtained as the intersection of the surface of a cone with a plane.



CONIC SECTIONS



CONIC SECTIONS (CONTD.....)

These are the loci of points moving in a plane such that the ratio of it's distances from a *fixed point* And a *fixed line* always remains constant.

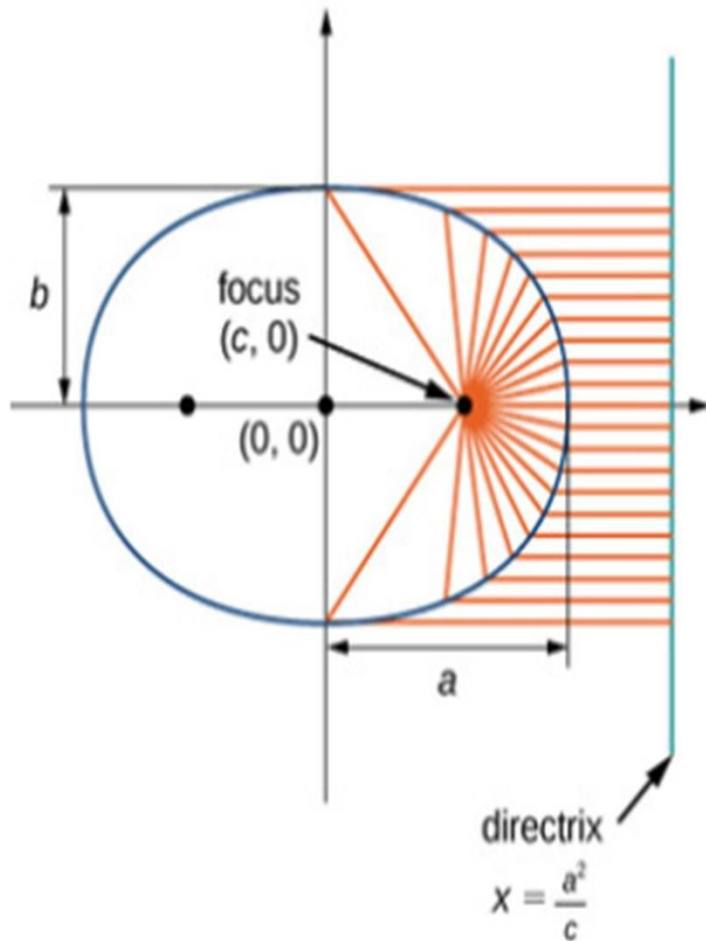
The Ratio is called **ECCENTRICITY**. (E)

- A) For Ellipse $E < 1$
- B) For Parabola $E = 1$
- C) For Hyperbola $E > 1$

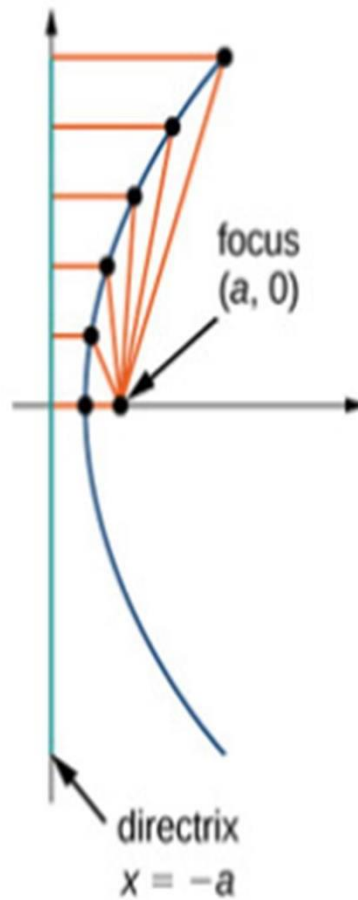
- The fixed point is called *focus*
- The fixed line is called *directrix*

Conic sections

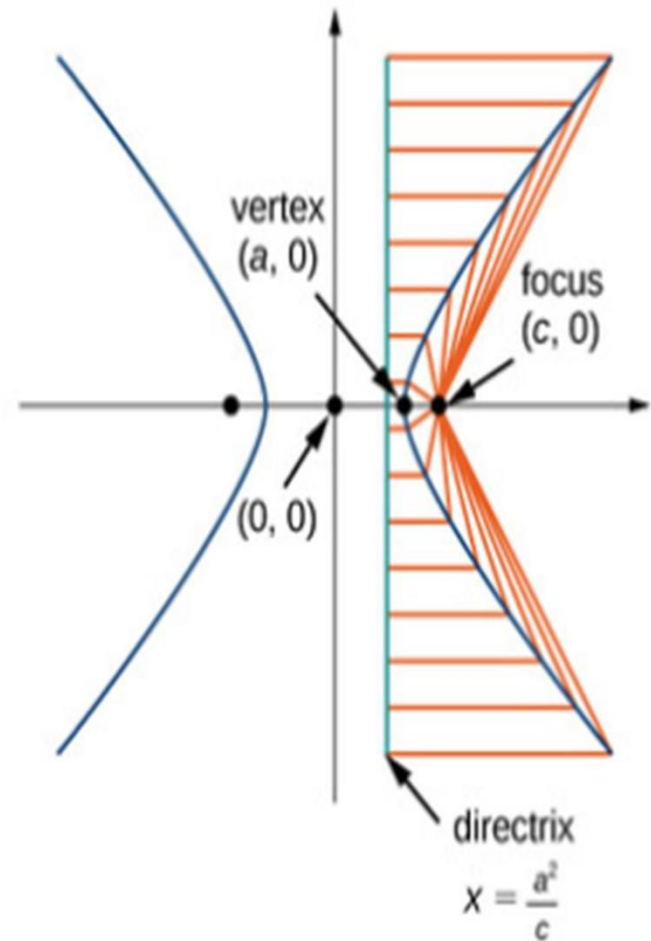
Ellipse



Parabola

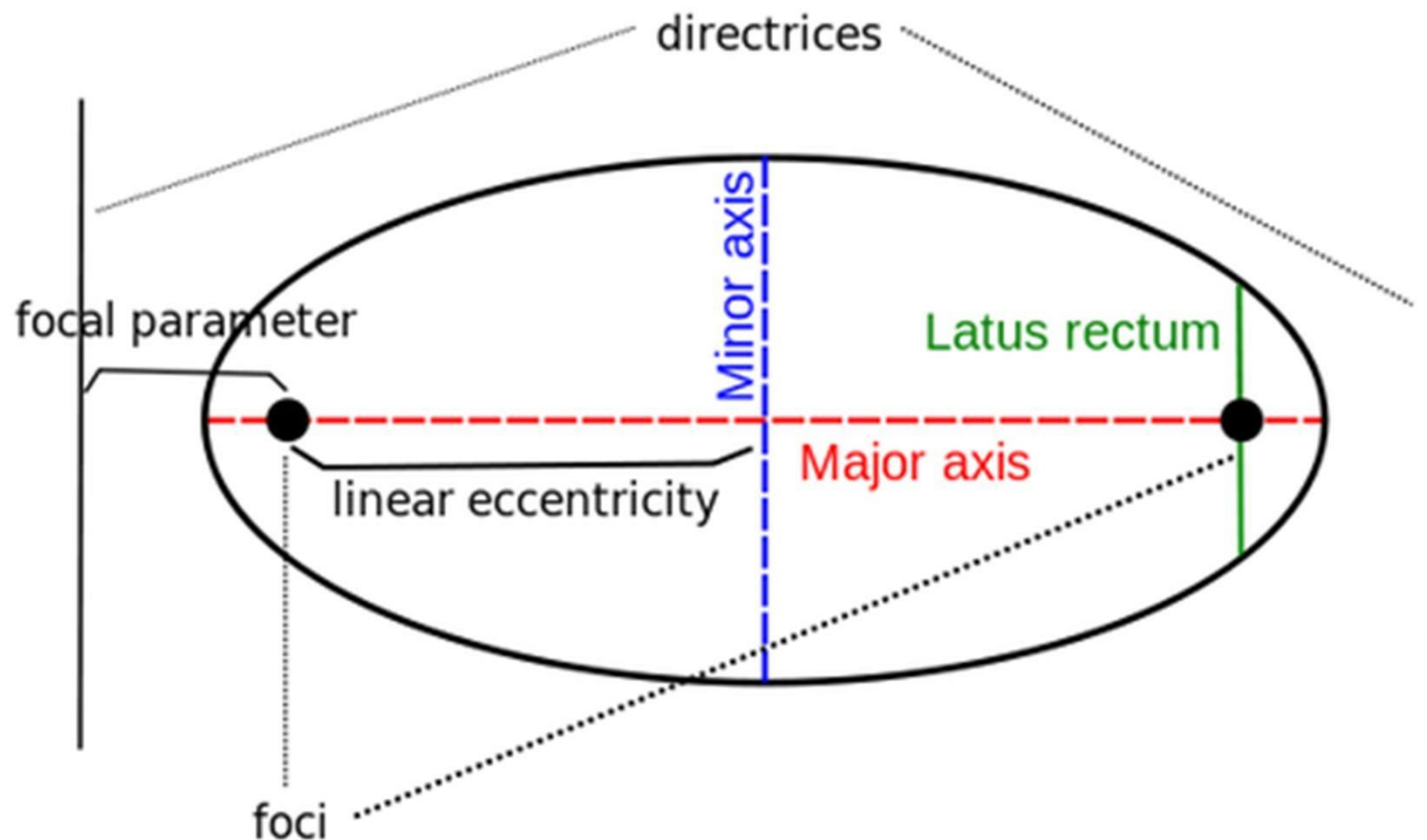


Hyperbola



ELLIPSE

It is a locus of a point moving in a plane such that the SUM of it's distances from TWO fixed points always remains constant. {And this sum equals to the length of major axis.} These TWO fixed points are FOCUS 1 & FOCUS 2



ELLIPSE (Contd.....)

Methods of Construction

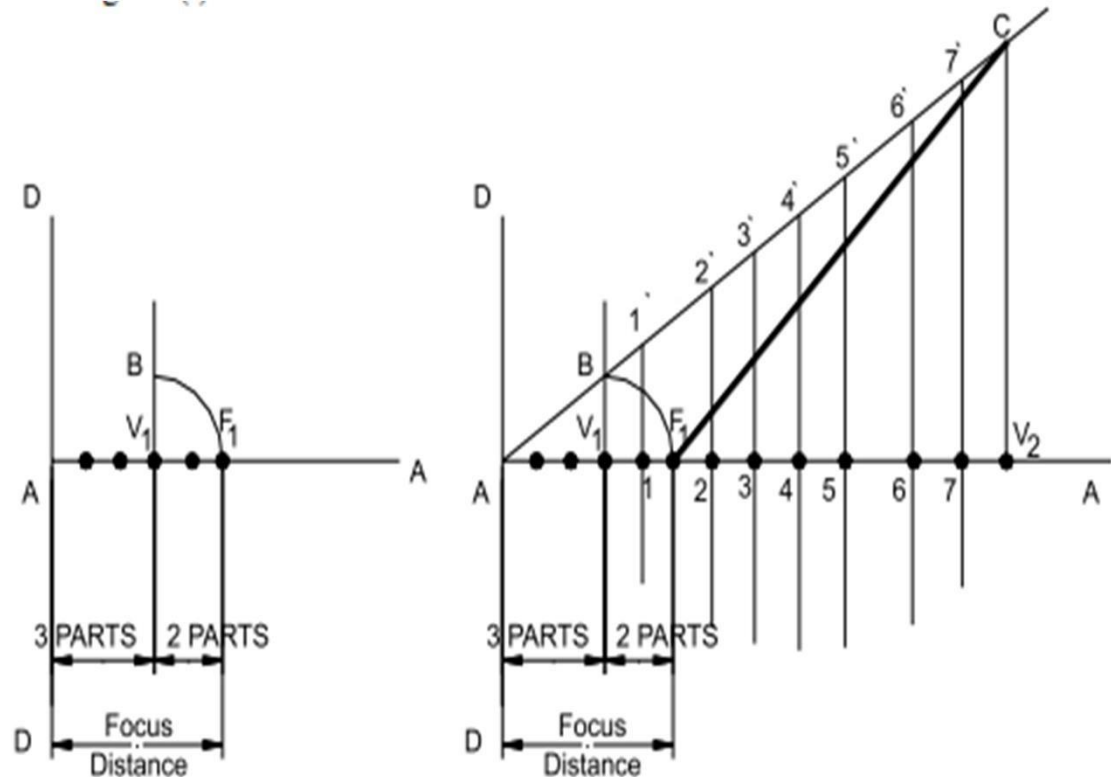
- 1. Basic Locus method (Directrix- Focus method)**
- 2. Concentric Circle Method**
- 3. Rectangle Method**
- 4. Oblong Method**
- 5. Arcs of Circle Method**
- 6. Rhombus Method**

PROBLEM 6:- POINT **F** IS 50 MM FROM A LINE **AD**. A POINT **P** IS MOVING IN A PLANE SUCH THAT THE **RATIO** OF IT'S DISTANCES FROM **F** AND LINE **AD** REMAINS CONSTANT AND EQUALS TO **2/3** DRAW LOCUS OF POINT **P**. { ECCENTRICITY = **2/3** }

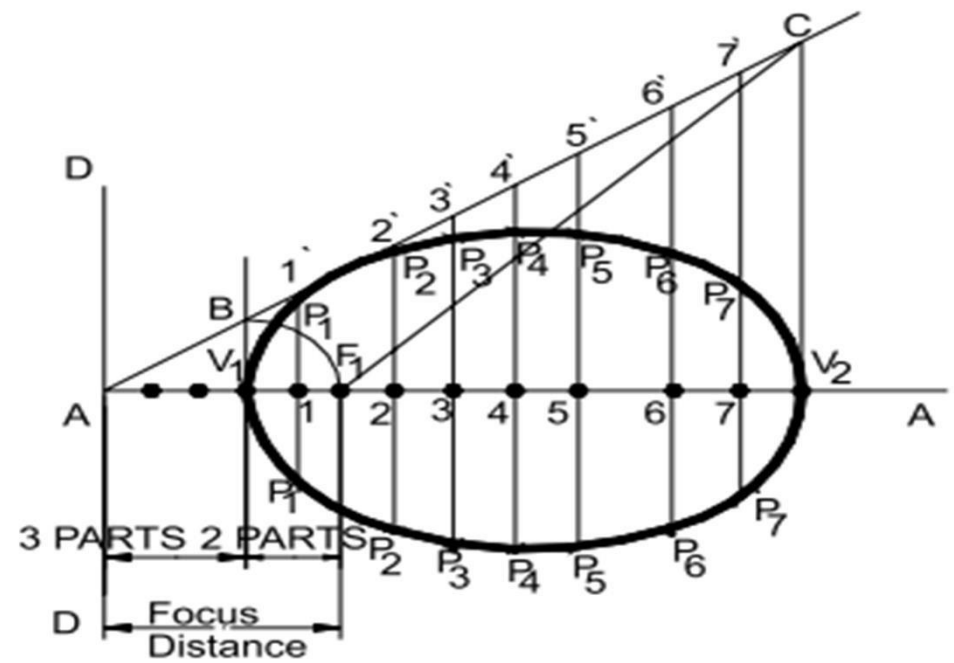
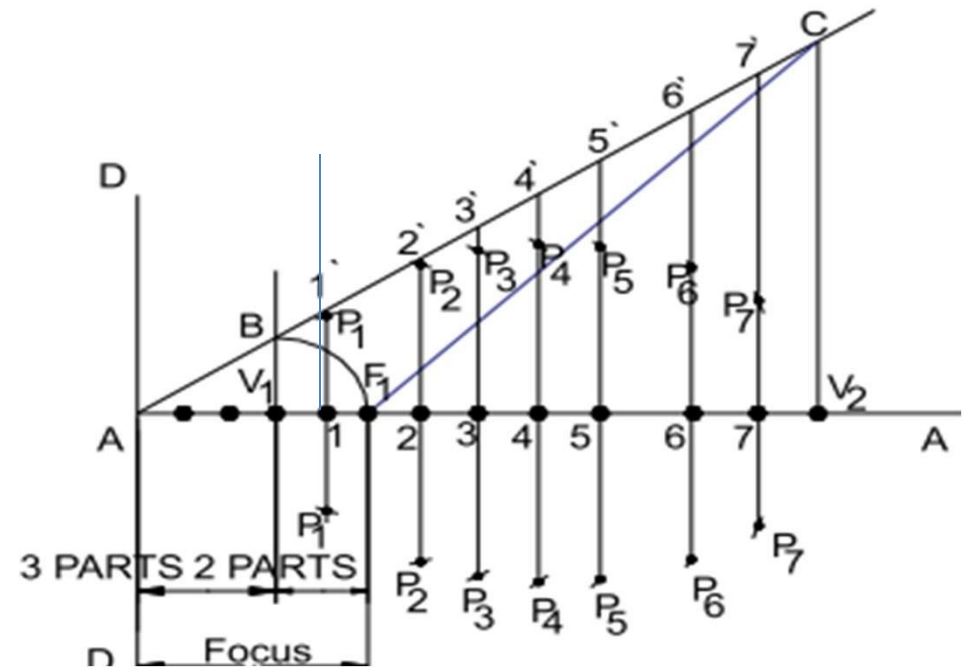
ELLIPSE

DIRECTRIX-FOCUS METHOD

- **STEPS:**
- Draw a vertical line **AD** and point **F₁** 50 mm from it.
- Divide 50 mm distance in 5 parts.
- Name 2nd part from **F₁** as **V₁**. It is 20mm and 30mm from **F₁** and **AD** line resp. It is first point giving ratio of it's distances from **F₁** and **AD** 2/3
- Draw a perpendicular line (any convenient length) at point **V₁** and taking radius as **V₁ F₁** and centre as **V₁**, draw an arc which cuts the perpendicular line at a point **B** such that **V₁ B = V₁ F₁**
- Join **A, B** and extend it conveniently.
- Draw a 45° line from the foci **F₁** such that it meets the extended **AB** line at a point **C**. Drop a vertical line from point **C** onto the axis line **AA**, which gives the second vertex **V₂**.

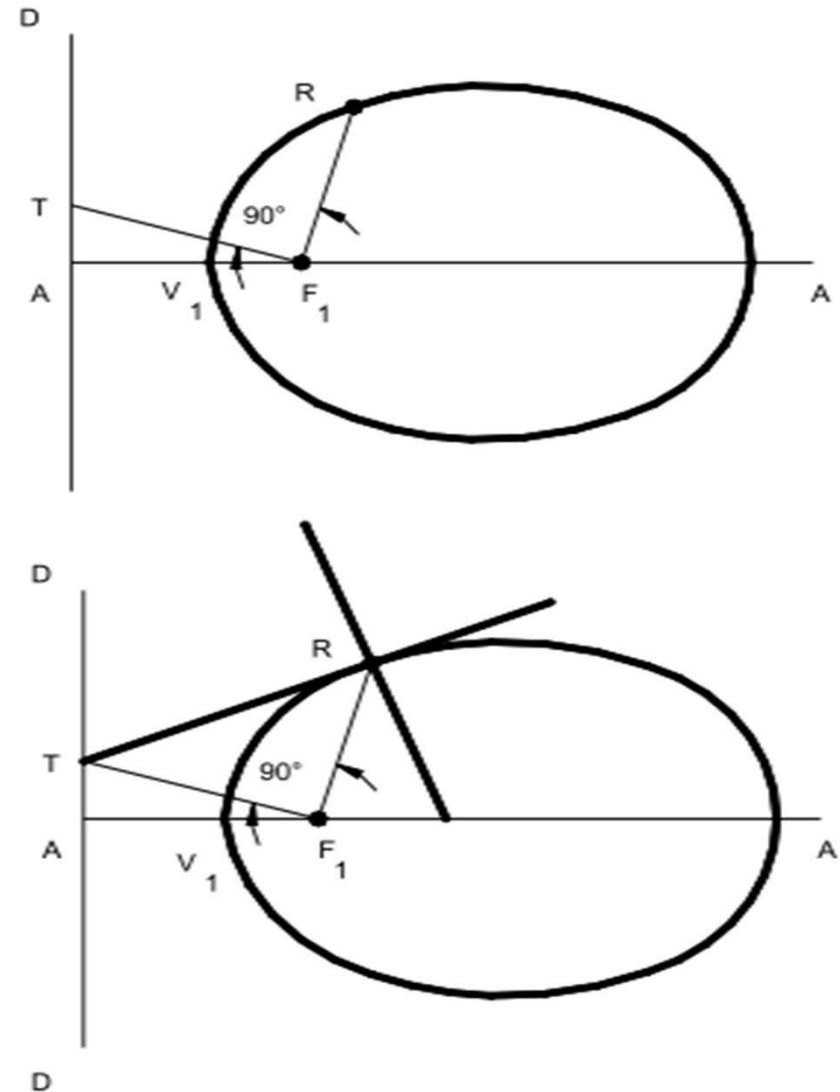


- Mark any number of points (which may or may not be equidistant) in between vertices V1 & V2 and name them 1, 2, 3,....
- Draw perpendicular lines at these divided points 1, 2, 3,.... such that they meet extends AB line at points 1', 2', 3',.... ,
- step 8: Take 1 – 1' as radius and centre as F1, cut the perpendicular line 1 – 1' on either side of the axis, to generate two points named P1 and P'1 on either side of the axis line.
- Step 9: Repeat step 8 taking (2 – 2'),(3 – 3'),(4 – 4')... as radius and centre as F1 only, cut the respective perpendicular lines (2 – 2'),(3 – 3'),(4 – 4')..., which generates points P2, P3, P4,... on either side of the axis line.
- Join all the points P1, P2, P3, P4,... including vertices V1 & V2 by smooth curves, which will give the required ellipse,



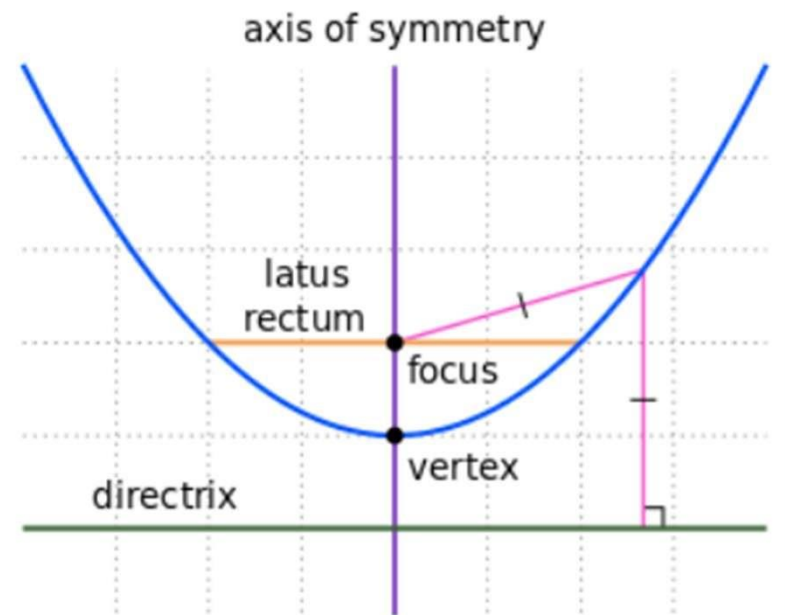
NORMAL AND TANGENT OF ELLIPSE

- Construct the conic section using the eccentricity method.
- Mark a point on the conic section with the given distance, where the normal and the tangent are required and name that point R. Join R to the focus point F₁.
- Draw a perpendicular line to RF₁, so that it touches the directrix at the point T
- Join points T, and R and extend it to get a tangent line.
- Draw a perpendicular line to the tangent TR to get the normal line, to the given conic section.



PARABOLA

- Parabola is as a conic section, created from the intersection of a right circular conical surface and a plane parallel to another plane that is tangential to the conical surface
- Parabola is the locus of points in that plane that are equidistant from both the directrix and the focus



Construction method

1. Basic Locus Method
(Directrix – focus)

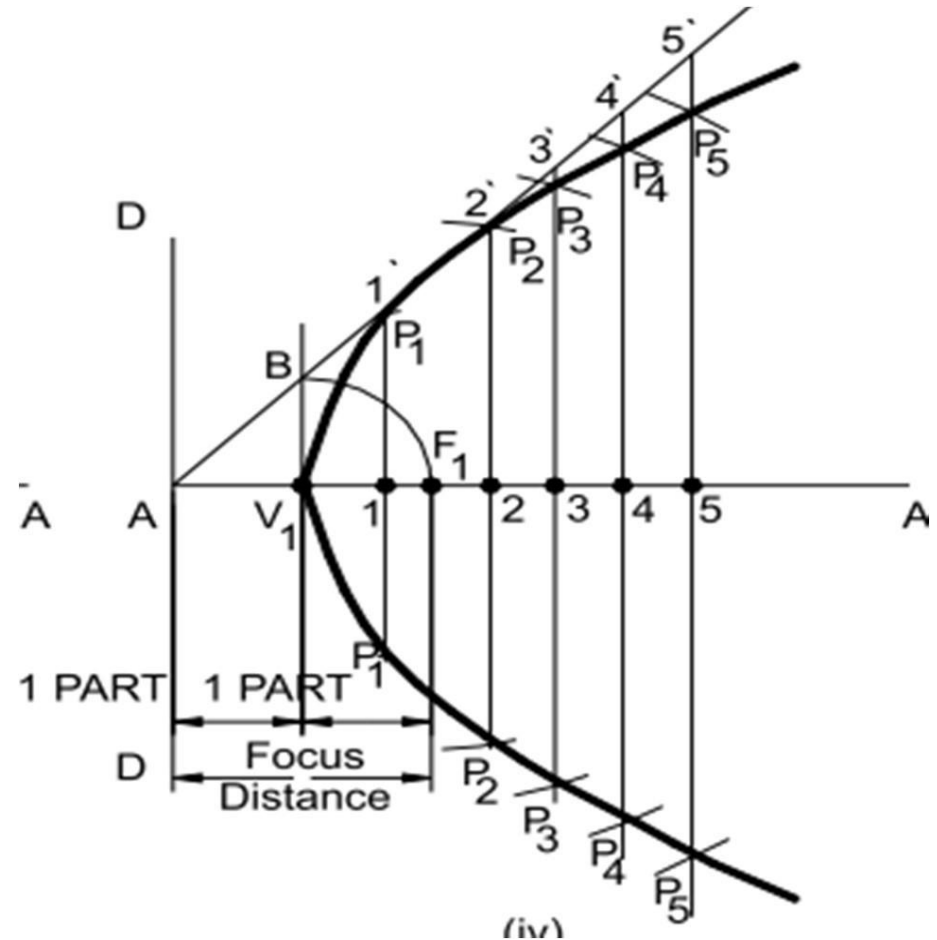
2. Rectangle Method

2 Method of Tangents
(Triangle Method)

PROBLEM 9: Point **F** is 50 mm from a vertical straight line **AD**. Draw locus of point **P**, moving in a plane such that it always remains equidistant from point **F** and line **AD**.

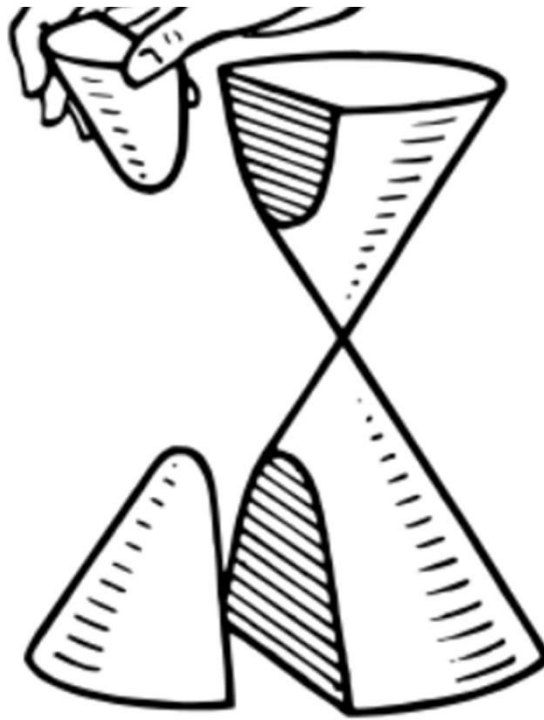
PARABOLA DIRECTRIX-FOCUS METHOD

- STEPS:
- Draw a vertical line AD and 50 mm from point F1 it.
- Divide AF1. Mark V1 at the center of AF1
- ☐ Mark any number of points (which may or may not be equidistant) and name them 1, 2, 3,....
- ☐ Draw perpendicular lines at these divided points 1, 2, 3,.... such that they meet extends AB line at points 1', 2', 3',....,
- ☐ Take 1 – 1' as radius and centre as F1, cut the perpendicular line 1 on either side of the axis, to generate two points named P1 and P'1 on either side of the axis line.
- ☐ Repeat step taking (2 – 2'), (3 – 3'), (4 – 4')... as radius and centre as F1 only, cut the respective perpendicular lines (2 2'), (3 – 3'), (4 – 4')..., which generates points P2, P3, P4,... on either side of the axis line
- ☐ Join all the points v1, P1, P2, P3, P4,... by smooth curves, which will give the required parabola.



HYPERBOLA

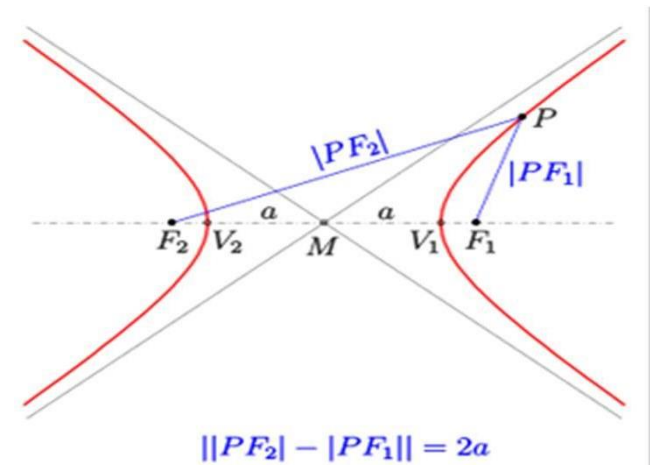
The hyperbola is one of the three kinds of conic section, formed by the intersection of a plane and a double cone.



HYPERBOLA

1. Rectangular Hyperbola (coordinates given)
- 2 Rectangular Hyperbola (P-V diagram - Equation given)
3. Basic Locus Method (Directrix – focus)

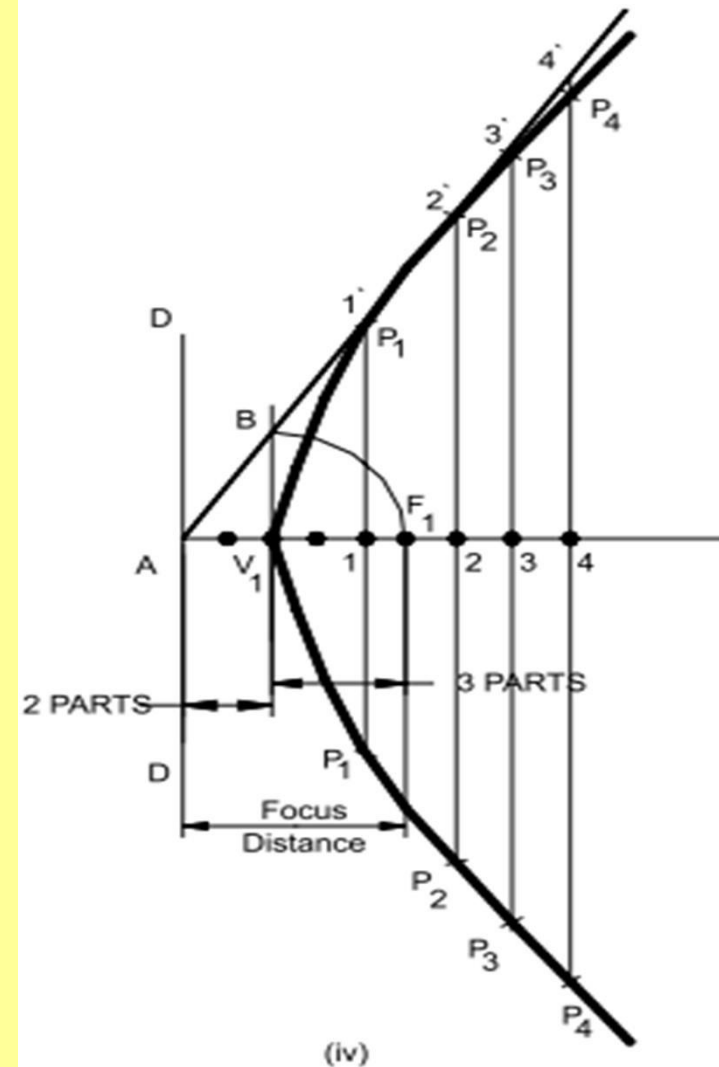
A **hyperbola** is a set of points, such that for any point P of the set, the absolute difference of the distances $|PF_1|$, $|PF_2|$, $|PF_1| - |PF_2|$ to two fixed points F_1 , F_2 , (the *foci*), is constant, usually denoted by $2a$,



PROBLEM 12:- POINT **F** IS 50 MM FROM A LINE **AB**. A POINT **P** IS MOVING IN A PLANE SUCH THAT THE RATIO OF IT'S DISTANCES FROM **F** AND LINE **AB** REMAINS CONSTANT AND EQUALS TO $\frac{3}{2}$ DRAW LOCUS OF POINT **P**. { ECCENTRICITY = $\frac{3}{2}$ }

- Draw a vertical directrix line **DD** and an axis line **AA** perpendicular to it, of convenient length. Mark the focus distance from the directrix on the axis line and name the point **F1**.
- Divide the line segment **AF1** into equal number of parts, such that the number of parts is equal to the sum of the numerator and the denominator of the eccentricity ratio, e.g., say, if the eccentricity ratio is $\frac{3}{2}$ then divide the line segment **AF1** into $(3 + 2 = 5)$ 5 parts.
- Use the eccentricity formula to locate the vertex **V1** on the axis line such that **V1 F1** is equal to 3 parts (numerator) and **AV1** is equal to 2 parts (denominator), among the five parts divided.
- Draw a perpendicular line at point **V1** of convenient length and taking radius as **V1 F1** and centre as **V1**, draw an arc which cuts the perpendicular line at a point named **B**, such that **V1 B = V1 F1**.
- Join **A, B** and extend it conveniently
- Mark any number of points (which may or may not be equidistant) to the right side of vertex **V1** and name them 1, 2, 3,...
- Draw perpendicular lines at these divided points 1, 2, 3,.... such that they meet, **AB** extend line at points 1', 2', 3',....,
- Take 1 – 1' as radius and centre as **F1**, cut the perpendicular line 1 – 1' on either side of the axis, to get 2 points named **P1** on either side of the axis line **AA**.

HYPERBOLA DIRECTRIX -FOCUS METHOD



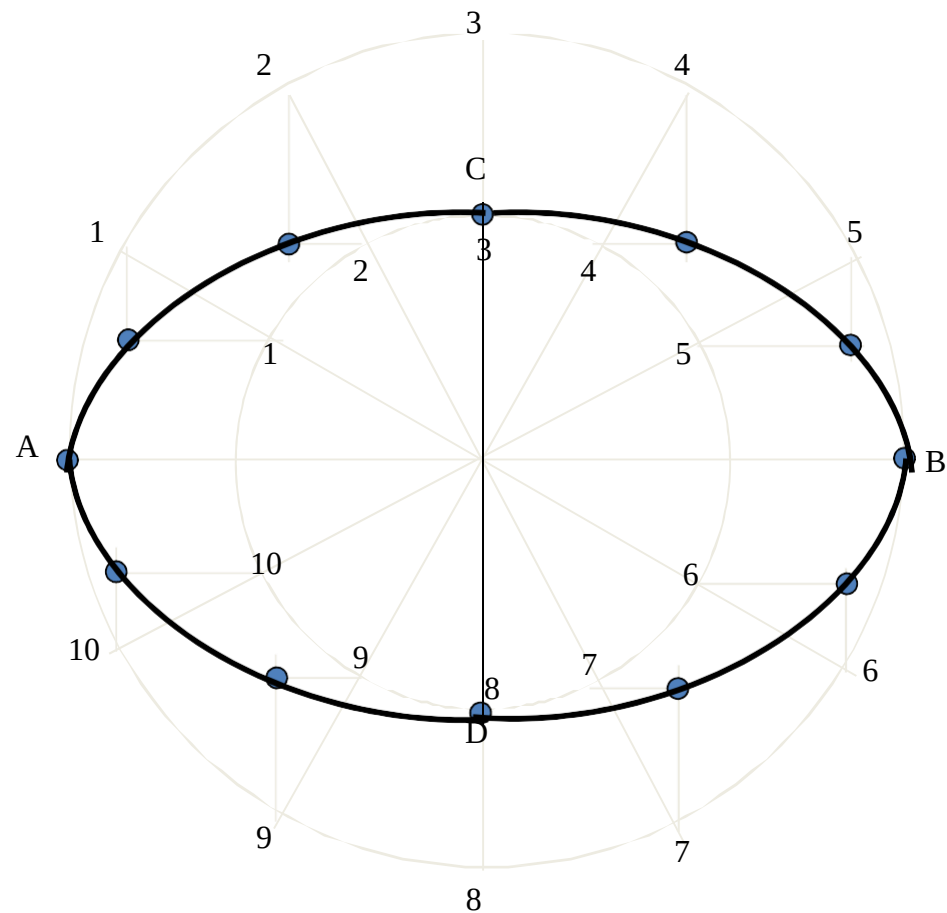
DIFFERENT METHODS FOR CONSTRUCTING ELLIPSE

Problem 1 :-

*Draw ellipse by concentric circle method.
Take major axis 100 mm and minor axis 70 mm long.*

Steps:

1. Draw both axes as perpendicular bisectors of each other & name their ends as shown.
2. Taking their intersecting point as a center, draw two concentric circles considering both as respective diameters.
3. Divide both circles in 12 equal parts & name as shown.
4. From all points of outer circle draw vertical lines downwards and upwards respectively.
5. From all points of inner circle draw horizontal lines to intersect those vertical lines.
6. Mark all intersecting points properly as those are the points on ellipse.
7. Join all these points along with the ends of both axes in smooth possible curve. It is required ellipse.



ELLIPSE **BY CONCENTRIC CIRCLE METHOD**

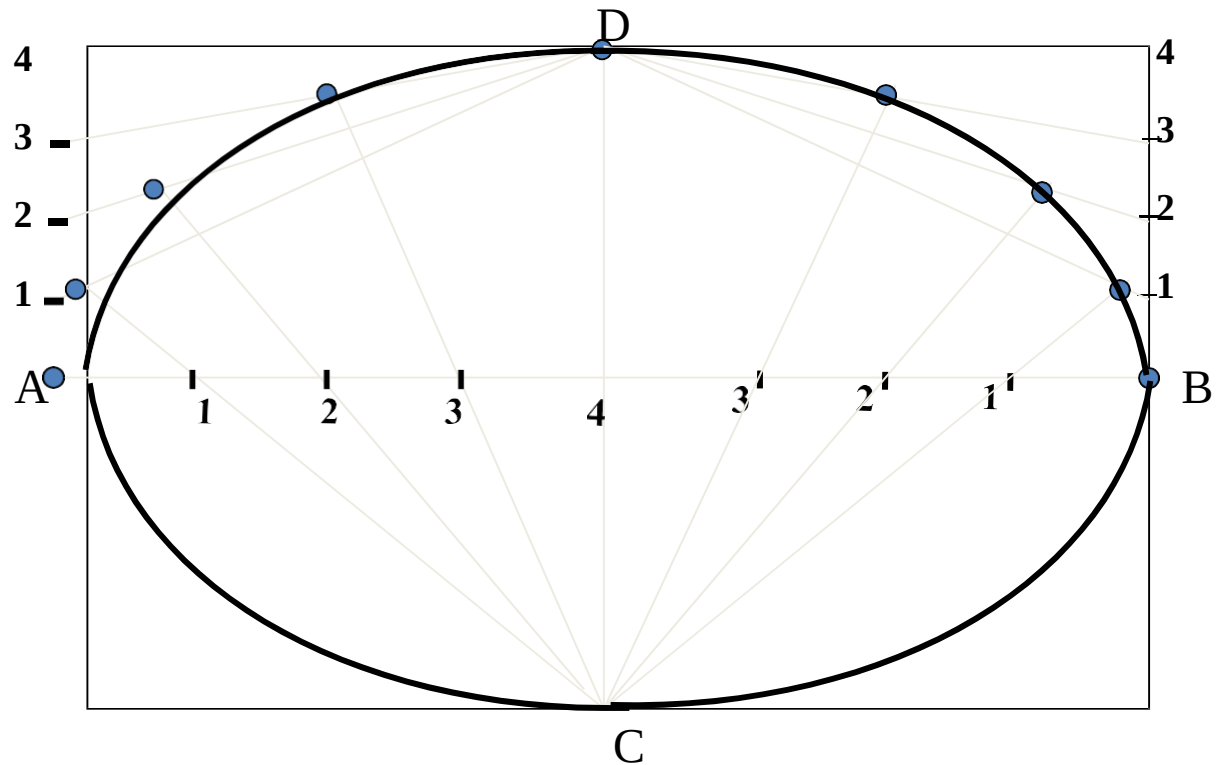
Steps:

- 1 Draw a rectangle taking major and minor axes as sides.
2. In this rectangle draw both axes as perpendicular bisectors of each other.
3. For construction, select upper left part of rectangle. Divide vertical small side and horizontal long side into same number of equal parts. (here divided in four parts)
4. Name those as shown..
5. Now join all vertical points 1,2,3,4, to the upper end of minor axis. And all horizontal points i.e. 1,2,3,4 to the lower end of minor axis.
6. Then extend C-1 line upto D-1 and mark that point. Similarly extend C-2, C-3, C-4 lines up to D-2, D-3, & D-4 lines.
7. Mark all these points properly and join all along with ends A and D in smooth possible curve. Do similar construction in right side part along with lower half of the rectangle. Join all points in smooth curve. It is required ellipse.

ELLIPSE
BY RECTANGLE METHOD

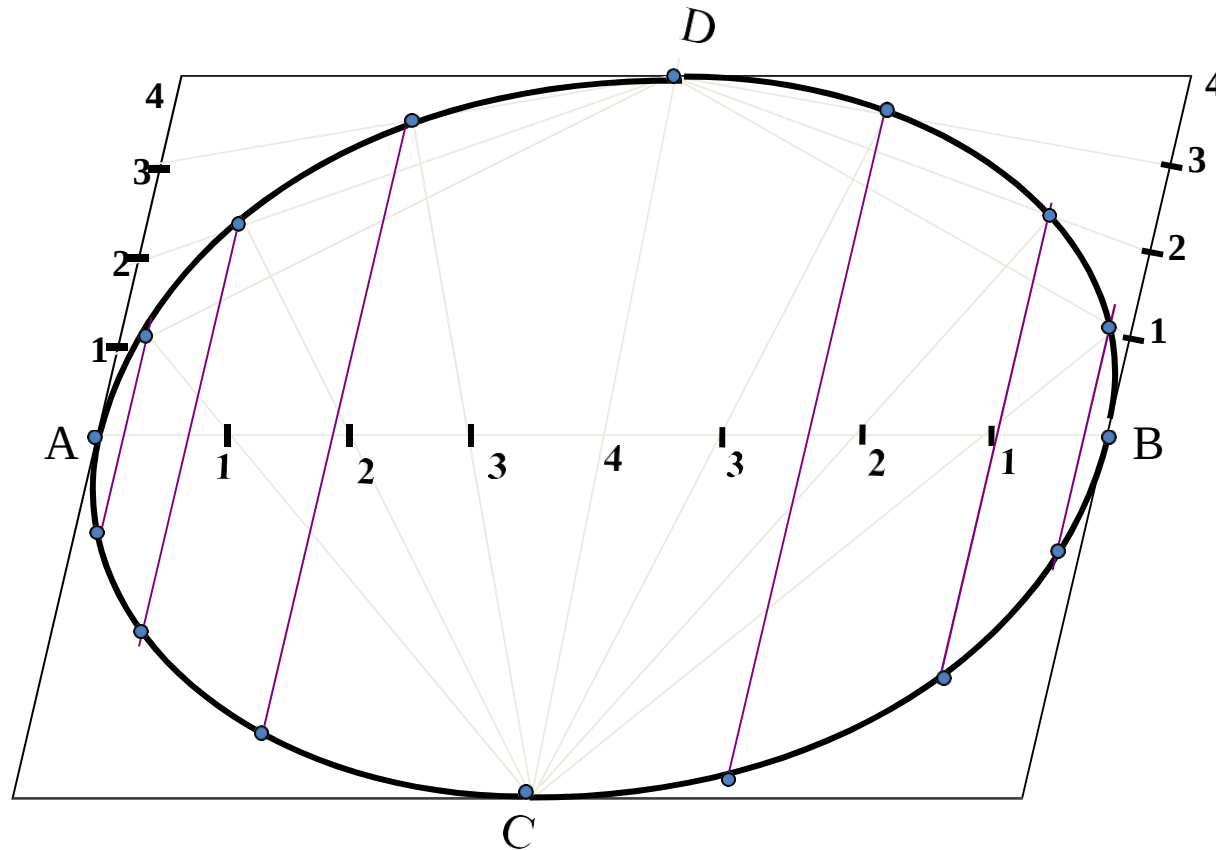
Problem 2

Draw ellipse by **Rectangle method**.
Take major axis 100 mm and minor axis 70 mm long.



Problem 3:- Draw ellipse by **Oblong method**.
Draw a parallelogram of 100 mm and 70 mm long
sides with included angle of 75° . Inscribe Ellipse in it.

ELLIPSE
BY OBLONG METHOD



PROBLEM 4.

MAJOR AXIS AB & MINOR AXIS CD ARE
100 AND 70MM LONG RESPECTIVELY
.DRAW ELLIPSE BY ARCS OF CIRCLES
METHOD.

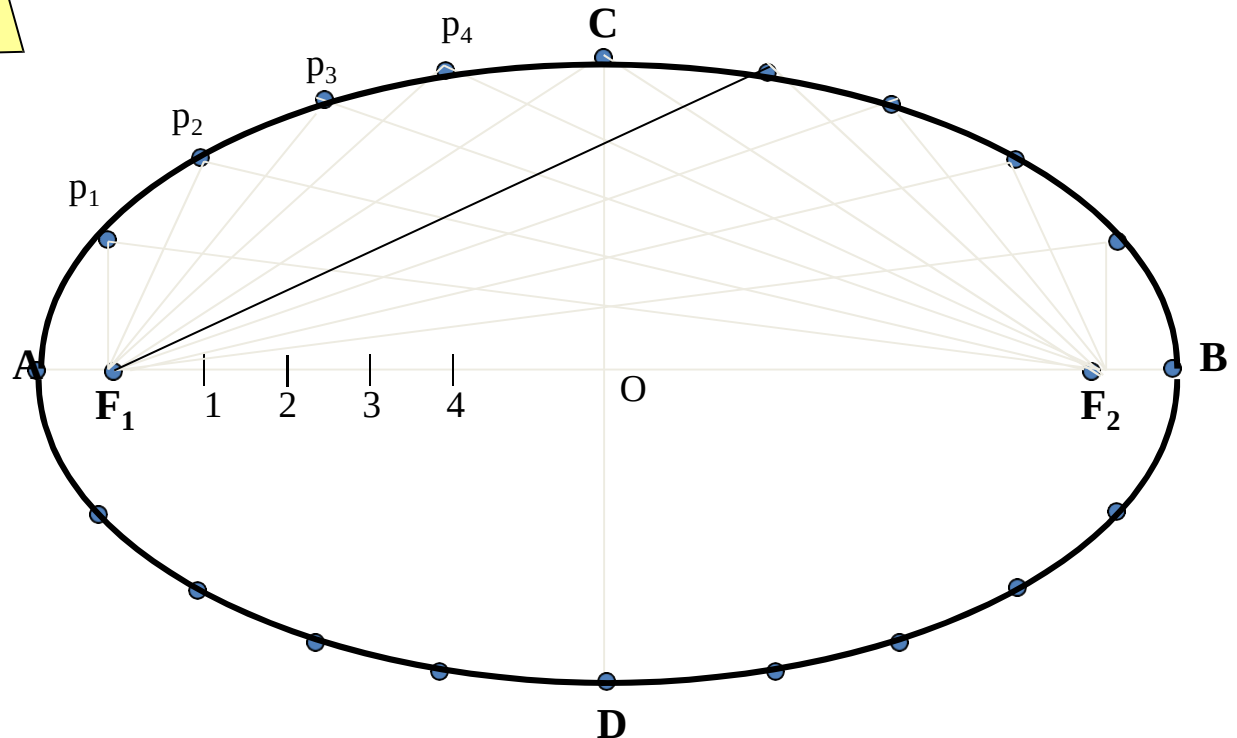
STEPS:

1. Draw both axes as usual. Name the ends & intersecting point
2. Taking AO distance i.e. half major axis, from C, mark F_1 & F_2 on AB. (focus 1 and 2.)
3. On line $F_1 - O$ taking any distance, mark points 1, 2, 3, & 4
4. Taking F_1 center, with distance A-1 draw an arc above AB and taking F_2 center, with B-1 distance cut this arc.
Name the point p_1
5. Repeat this step with same centers but
taking now A-2 & B-2 distances for drawing arcs. Name the point p_2
6. Similarly get all other P points.
With same steps positions of P can be
located below AB.
7. Join all points by smooth curve to get
an ellipse/

ELLIPSE

BY ARCS OF CIRCLE METHOD

As per the definition Ellipse is locus of point P moving in a plane such that the **SUM** of its distances from two fixed points (F_1 & F_2) remains constant and equals to the length of major axis AB. (Note $A . 1 + B . 1 = A . 2 + B . 2 = AB$)



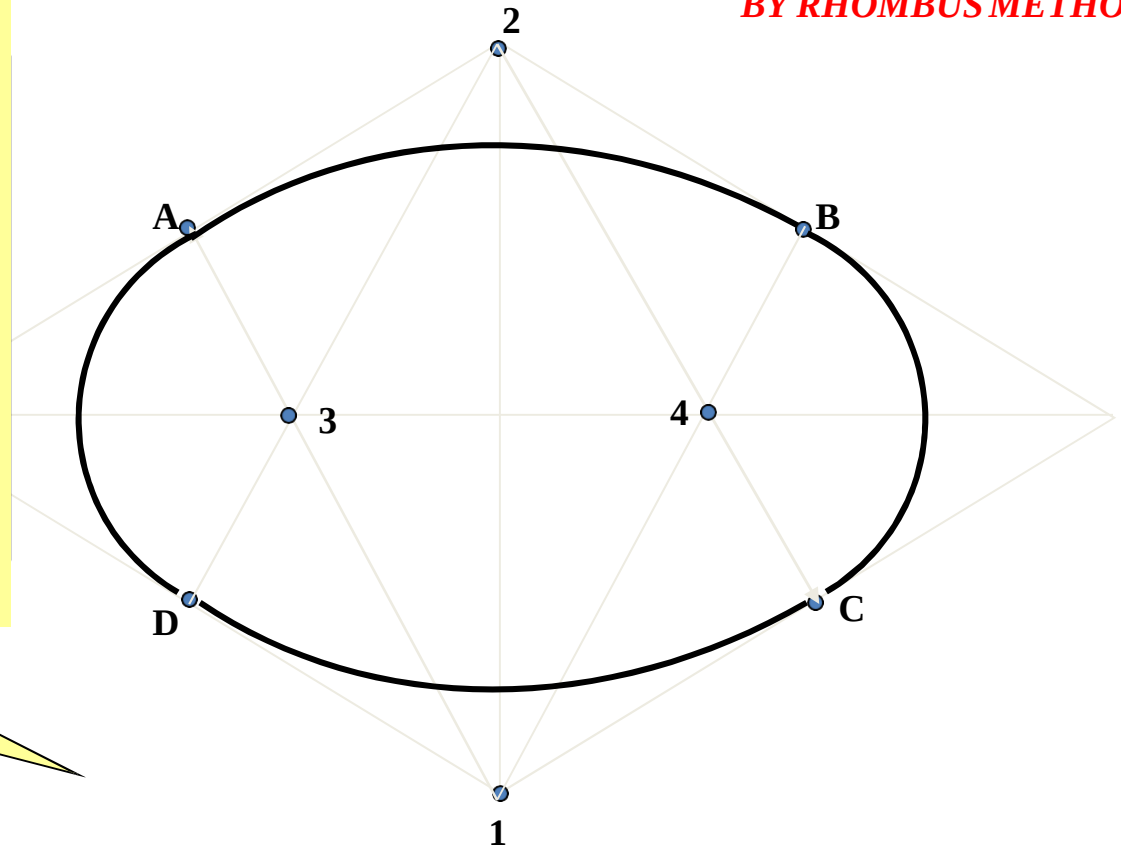
PROBLEM 5.

DRAW RHOMBUS OF 100 MM & 70 MM LONG
DIAGONALS AND INSCRIBE AN ELLIPSE IN IT.

STEPS:

1. Draw rhombus of given dimensions.
2. Mark mid points of all sides & name Those A,B,C,& D
3. Join these points to the ends of smaller diagonals.
4. Mark points 1,2,3,4 as four centers.
5. Taking 1 as center and 1-A radius draw an arc AB.
6. Take 2 as center draw an arc CD.
7. Similarly taking 3 & 4 as centers and 3-D radius draw arcs DA & BC.

ELLIPSE
BY RHOMBUS METHOD

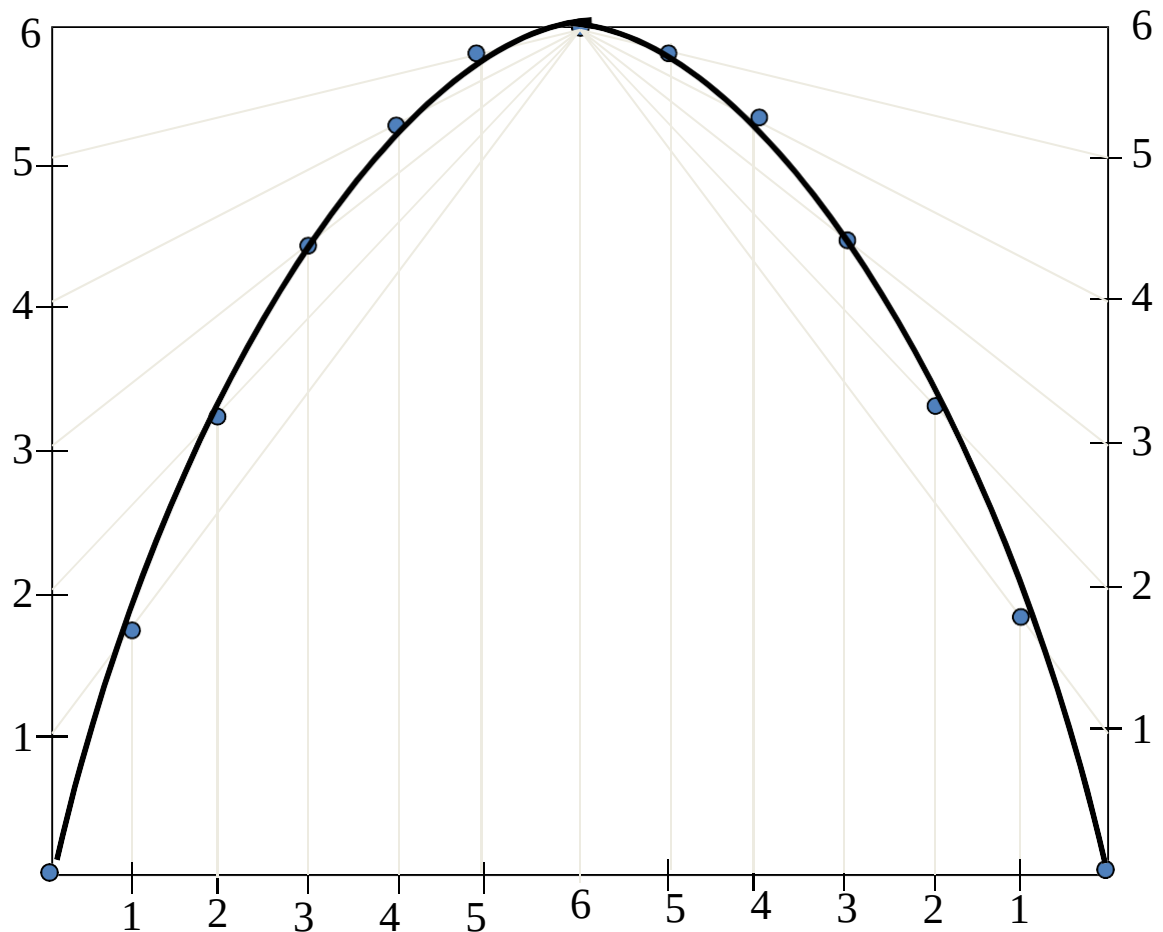


PROBLEM 7: A BALL THROWN IN AIR ATTAINS 100 M HEIGHT AND COVERS HORIZONTAL DISTANCE 150 M ON GROUND.
Draw the path of the ball (projectile)-

PARABOLA **RECTANGLE METHOD**

STEPS:

1. Draw rectangle of above size and divide it in two equal vertical parts
 2. Consider left part for construction. Divide height and length in equal number of parts and name those 1,2,3,4,5 & 6
 3. Join vertical 1,2,3,4,5 & 6 to the top center of rectangle
 4. Similarly draw upward vertical lines from horizontal 1,2,3,4,5. And wherever these lines intersect previously drawn inclined lines in sequence Mark those points and further join in smooth possible curve.
 5. Repeat the construction on right side rectangle also. Join all in sequence.
- This locus is Parabola.**

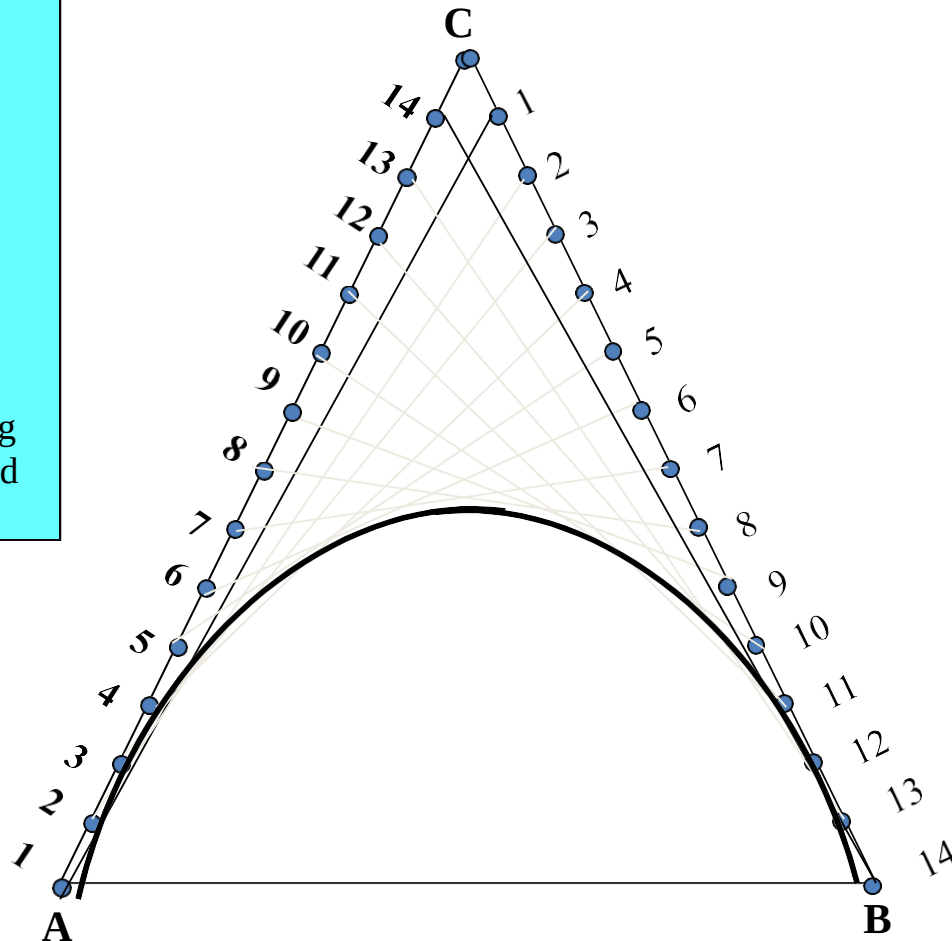


Problem no.8: Draw an isosceles triangle of 100 mm long base and 110 mm long altitude. Inscribe a parabola in it by method of tangents.

Solution Steps:

1. Construct triangle as per the given dimensions.
2. Divide its both sides into same no. of equal parts.
3. Name the parts in ascending and descending manner, as shown.
4. Join 1-1, 2-2, 3-3 and so on.
5. Draw the curve as shown i.e. tangent to all these lines. The above all lines being tangents to the curve, it is called method of tangents.

PARABOLA
METHOD OF TANGENTS

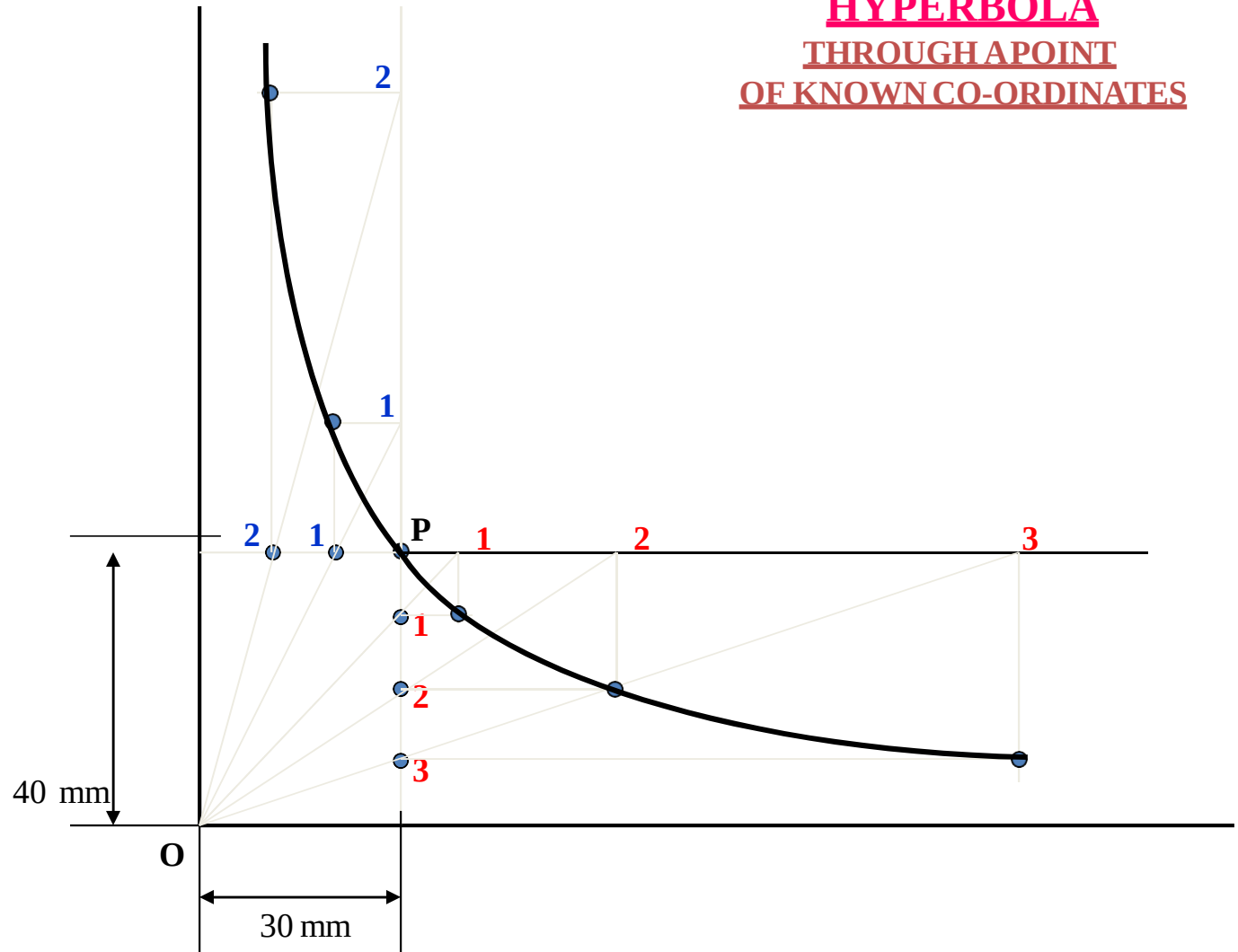


Problem No.10: Point P is 40 mm and 30 mm from horizontal and vertical axes respectively. Draw Hyperbola through it.

Solution Steps:

1. Extend horizontal line from P to right side.
2. Extend vertical line from P upward.
3. On horizontal line from P, mark some points taking any distance and name them after P-1, 2,3,4 etc.
- 4 Join 1-2-3-4 points to pole O. Let them cut part [P-B] also at 1,2,3,4 points.
5. From horizontal 1,2,3,4 draw vertical lines downwards and 6. From vertical 1,2,3,4 points [from P-B] draw horizontal lines.
7. Line from 1 horizontal and line from 1 vertical will meet at P_1 . Similarly mark P_2, P_3, P_4 points.
8. Repeat the procedure by marking four points on upward vertical line from P and joining all those to pole
9. Name this points P_6, P_7, P_8 etc. and join them by smooth curve.

HYPERBOLA
THROUGH A POINT
OF KNOWN CO-ORDINATES

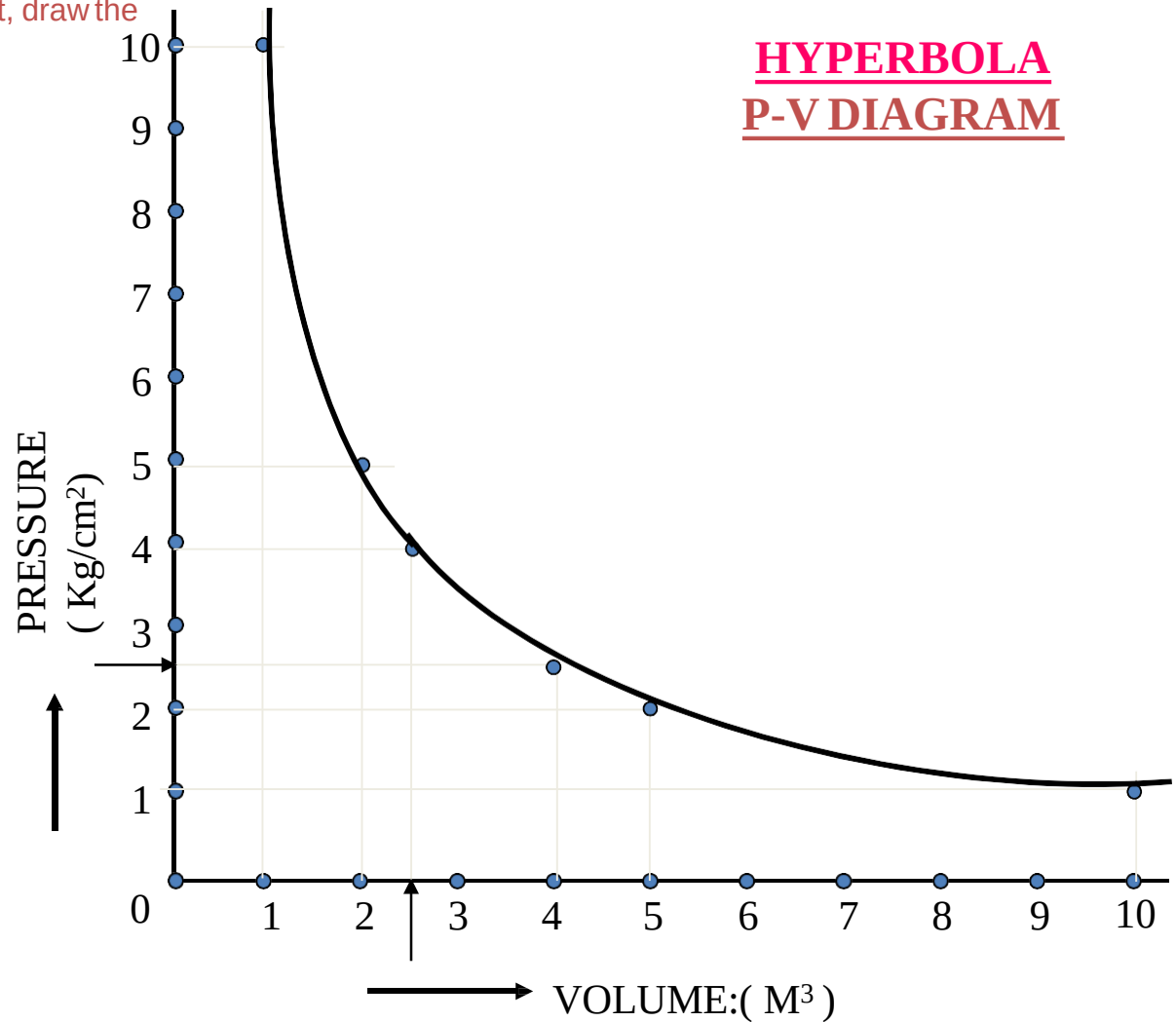


Problem no.11: A sample of gas is expanded in a cylinder from 10 unit pressure to 1 unit pressure. Expansion follows law $PV = \text{Constant}$. If initial volume being 1 unit, draw the curve of expansion. Also Name the curve.

Form a table giving few more values of P & V

$P \times V = C$		
10	1	= 10
5	2	= 10
4	2.5	= 10
2.5	4	= 10
2	5	= 10
1	10	= 10

Now draw a Graph of Pressure against Volume.
It is a PV Diagram and it is Hyperbola.
Take pressure on vertical axis and Volume on horizontal axis.

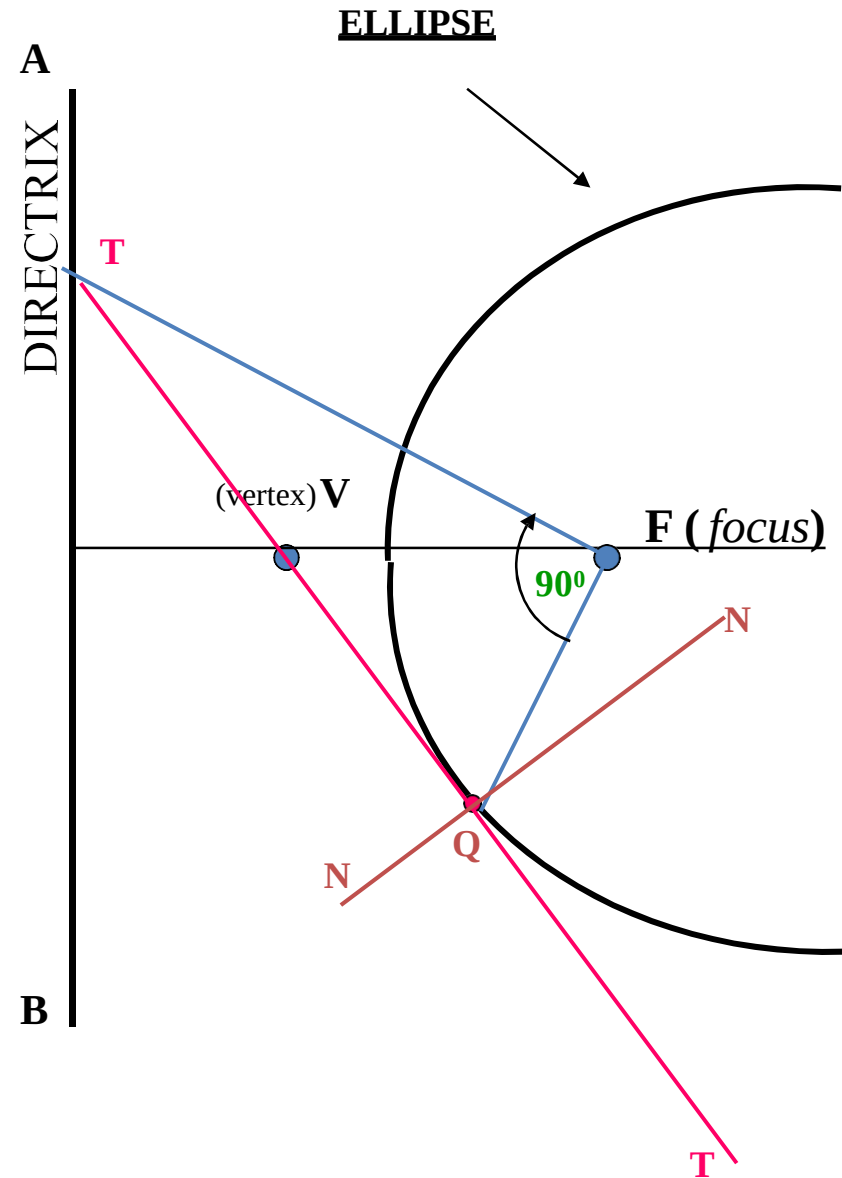


ELLIPSE TANGENT & NORMAL

Problem 14:

**TO DRAW TANGENT & NORMAL
TO THE CURVE
FROM A GIVEN POINT (Q)**

1. JOIN POINT **Q** TO **F**.
2. CONSTRUCT 90° ANGLE WITH THIS LINE AT POINT **F**
3. EXTEND THE LINE TO MEET DIRECTRIX AT **T**
4. JOIN THIS POINT TO **Q** AND EXTEND. THIS IS TANGENT TO ELLIPSE FROM **Q**
5. TO THIS TANGENT DRAW PERPENDICULAR LINE FROM **Q**. IT IS NORMAL TO CURVE.

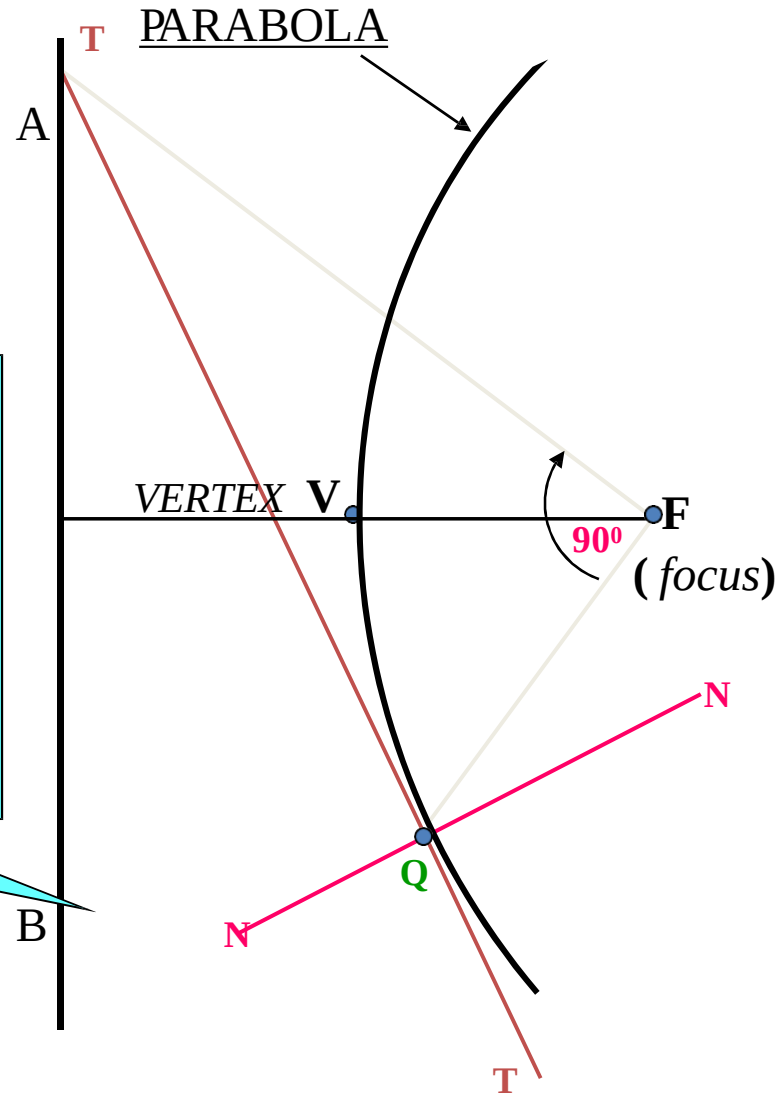


PARABOLA TANGENT & NORMAL

Problem 15:

**TO DRAW TANGENT & NORMAL
TO THE CURVE
FROM A GIVEN POINT (Q)**

1. JOIN POINT Q TO F.
2. CONSTRUCT 90° ANGLE WITH THIS LINE AT POINT F
3. EXTEND THE LINE TO MEET DIRECTRIX AT T
4. JOIN THIS POINT TO Q AND EXTEND. THIS IS TANGENT TO THE CURVE FROM Q
5. TO THIS TANGENT DRAW PERPENDICULAR LINE FROM Q. IT IS NORMAL TO CURVE.

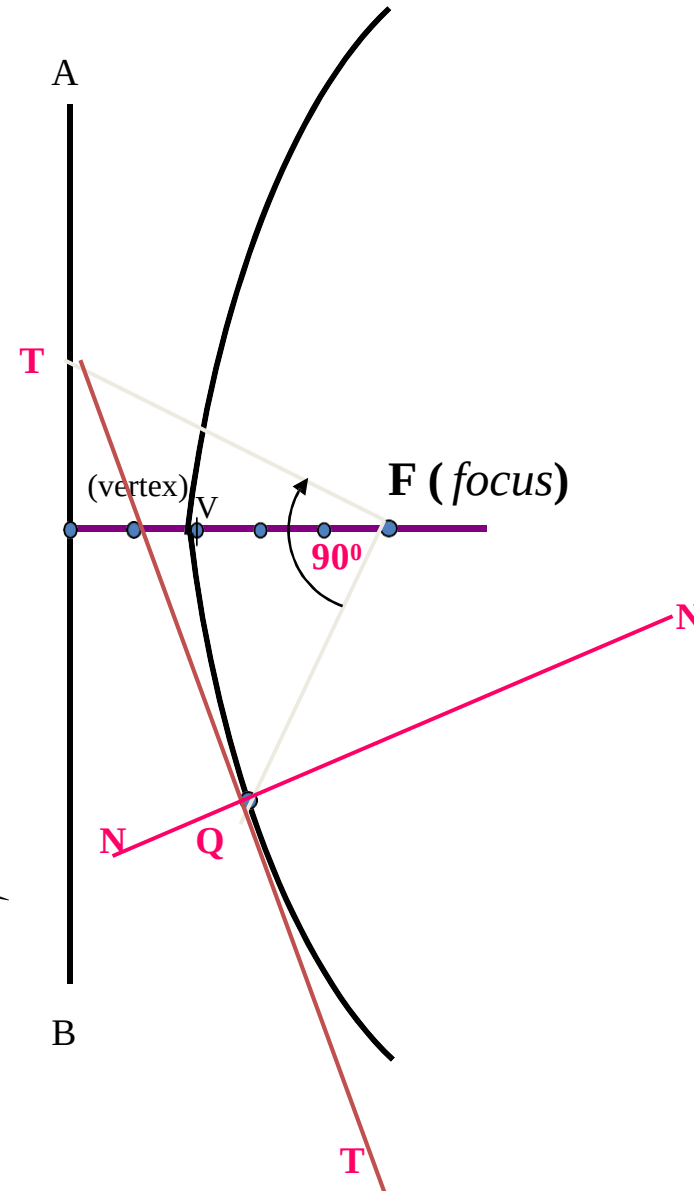


HYPERBOLA TANGENT & NORMAL

Problem 16

**TO DRAW TANGENT & NORMAL
TO THE CURVE
FROM A GIVEN POINT (Q)**

1. JOIN POINT **Q** TO **F**.
2. CONSTRUCT 90° ANGLE WITH THIS LINE AT POINT **F**
3. EXTEND THE LINE TO MEET DIRECTRIX AT **T**
4. JOIN THIS POINT TO **Q** AND EXTEND. THIS IS TANGENT TO CURVE FROM **Q**
5. TO THIS TANGENT DRAW PERPENDICULAR LINE FROM **Q**. IT IS NORMAL TO CURVE.

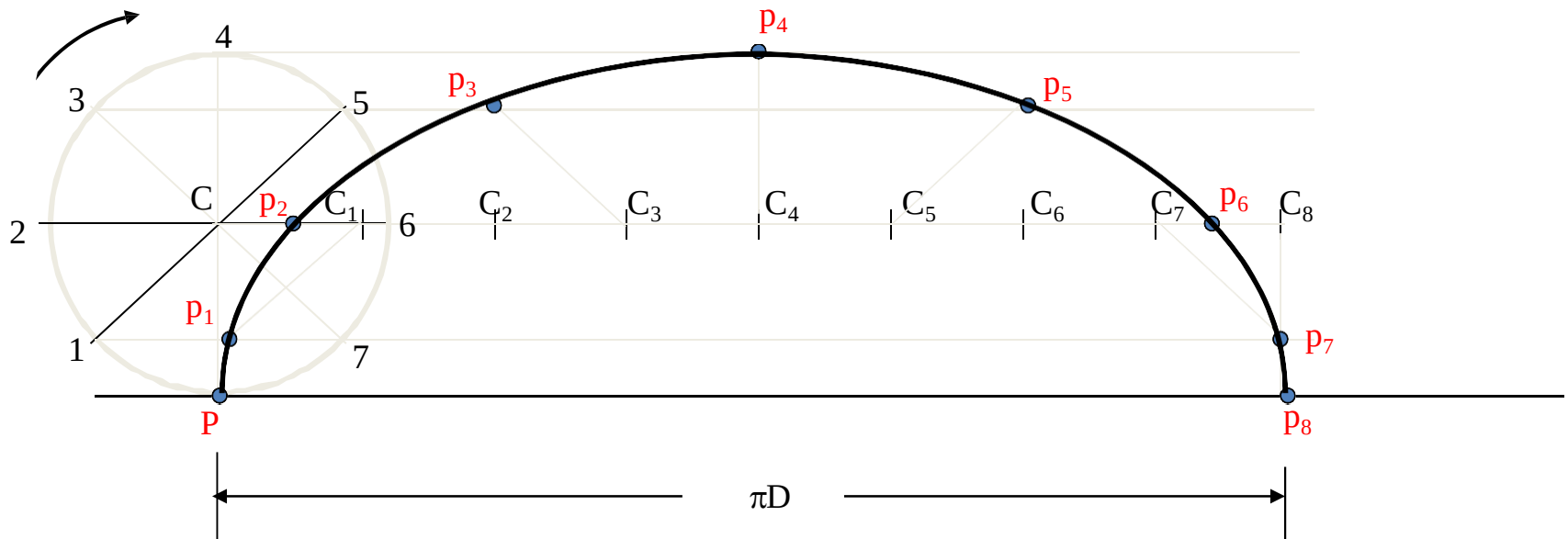


CYCLOID

A **cycloid** is the curve traced by a point on the rim of a circular wheel as the wheel rolls along a straight line without slipping.

PROBLEM 22: DRAW LOCUS OF A POINT ON THE PERIPHERY OF A CIRCLE WHICH ROLLS ON STRAIGHT LINE PATH. Take Circle diameter as 50 mm

CYCLOID



Steps

- 1) From center **C** draw a horizontal line equal to πD distance.
- 2) Divide πD distance into **8** number of equal parts and name them **C1, C2, C3** etc.
- 3) Divide the circle also into 8 number of equal parts and in clock wise direction, after **P** name **1, 2, 3** up to **8**.
- 4) From all these points on circle **draw horizontal lines**. (parallel to locus of **C**)
- 5) With a fixed distance **C-P** in compass, **C1** as center, mark a point on horizontal line from 1. Name it **P1**.
- 6) Repeat this procedure from **C2, C3, C4** upto **C8** as centers. Mark points **P2, P3, P4, P5** up to **P8** on the horizontal lines drawn from **2, 3, 4, 5, 6, 7** respectively.
- 7) Join all these points by curve. **It is Cycloid.**

SPIRAL

A curve on a plane that winds around a fixed center point at a continuously increasing or decreasing distance from the point.

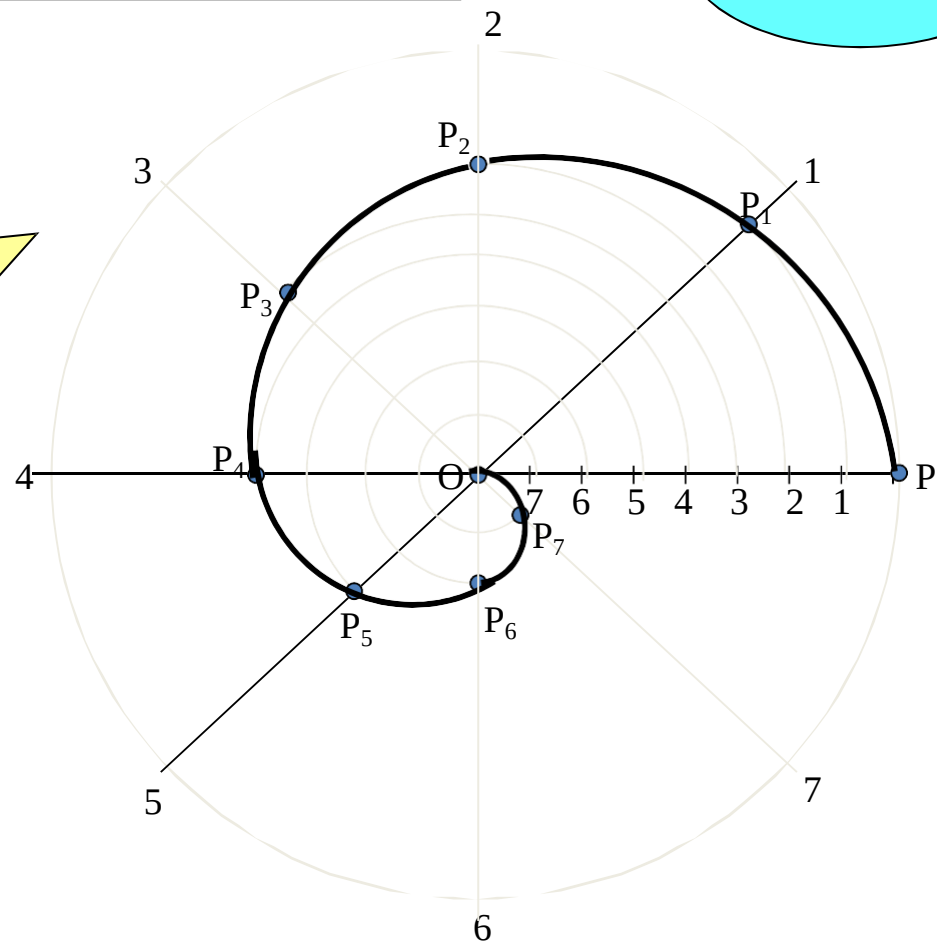
Problem 27: Draw a spiral of one convolution. Take distance PO 40 mm.

IMPORTANT APPROACH FOR CONSTRUCTION!
FIND TOTAL ANGULAR AND TOTAL LINEAR DISPLACEMENT
AND DIVIDE BOTH IN TO SAME NUMBER OF EQUAL PARTS.

SPIRAL

Solution Steps

1. With PO radius draw a circle and divide it in EIGHT parts. Name those 1,2,3,4, etc. up to 8
2. Similarly divided line PO also in EIGHT parts and name those 1,2,3,-- as shown.
3. Take o-1 distance from op line and draw an arc up to O1 radius vector. Name the point P_1
4. Similarly mark points P_2, P_3, P_4 up to P_8
And join those in a smooth curve.
It is a SPIRAL of one convolution.



IMPORTANT APPROACH FOR CONSTRUCTION!

FIND TOTAL ANGULAR AND TOTAL LINEAR DISPLACEMENT
AND DIVIDE BOTH IN TO SAME NUMBER OF EQUAL PARTS.

Problem 28

Point P is 80 mm from point O. It starts moving towards O and reaches it in two revolutions around it. Draw locus of point P (To draw a Spiral of TWO convolutions).

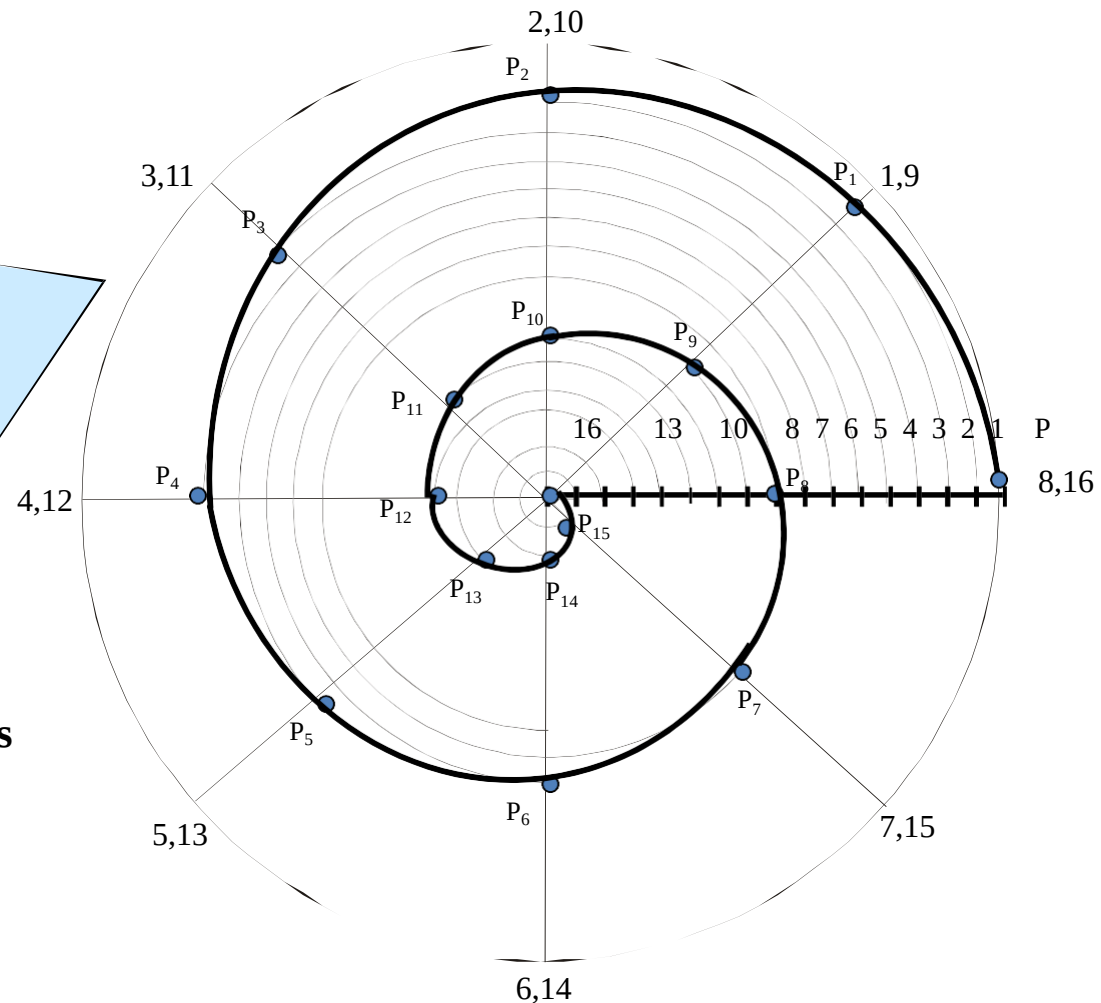
SPIRAL of two convolutions

SOLUTION STEPS:

Total angular displacement here
is two revolutions And
Total Linear displacement here
is distance PO.

Just divide both in same parts i.e.
Circle in EIGHT parts.
(means total angular displacement
in SIXTEEN parts)

Divide PO also in SIXTEEN parts.
Rest steps are similar to the previous
problem.



INVOLUTE

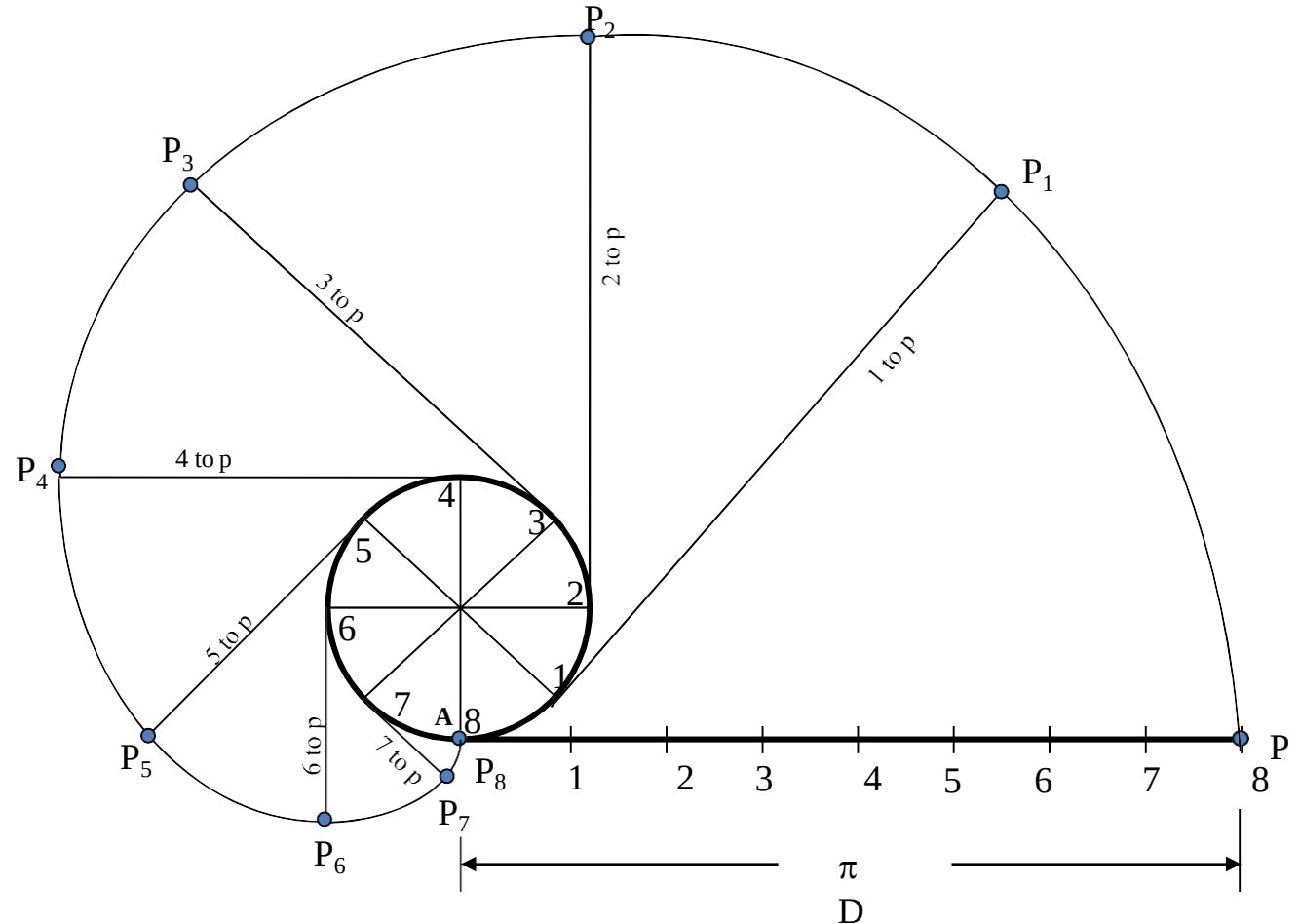
An involute (also known as an evolvent) is a particular type of curve that is dependent on another shape or curve. An involute of a curve is the locus of a point on a piece of taut string as the string is either unwrapped from or wrapped around the curve.

INVOLUTE OF A CIRCLE

Steps

- 1) Point or end P of string AP is exactly πD distance away from A. Means if this string is wound round the circle, it will completely cover given circle. P will meet A after winding.
- 2) Divide πD (AP) distance into 8 number of equal parts.
- 3) Divide circle also into 8 number of equal parts.
- 4) Name after A, 1, 2, 3, 4, etc. up to 8 on πD line AP as well as on circle (in anticlockwise direction).
- 5) To radius C-1, C-2, C-3 up to C-8 draw tangents (from 1, 2, 3, 4, etc. to circle).
- 6) Take distance 1 to P in compass and mark it on tangent from point 1 on circle (means one division less than distance AP).
- 7) Name this point P1
- 8) Take 2-P distance in compass and mark it on the tangent from point 2. Name it point P2.
- 9) Similarly take 3 to P, 4 to P, 5 to P up to 7 to P distance in compass and mark on respective tangents and locate P3, P4, P5 up to P8 (i.e. A) points and join them in smooth curve it is an INVOLUTE of a given circle.

Problem no 17: Draw Involute of a circle.
String length is equal to the circumference of circle.



Reference Youtube Videos

- ELLIPSE DRAWING

<https://www.youtube.com/watch?v=qkPZgVbtiHE>

- PARABOLA DRAWING

- <https://www.youtube.com/watch?v=ZlekZGPfbo8>

- Hyperbola Drawing:

- <https://www.youtube.com/watch?v=dcaGNfplUbU>

Reference Youtube Videos

- Cycloid drawing

<https://www.youtube.com/watch?v=UiNKuPztBfg>

- Involute Drawing

<https://www.youtube.com/watch?v=WmhOVyQVveQ>